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Software Effort Estimation using Ensemble Learning Methods

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I. Introduction

- Estimating software effort is a critical component of project management.
- Traditional estimation methods often struggle with inaccuracy due to project complexity.
- Machine Learning (ML) offers promising solutions for predicting development effort.
- Objective: This study focuses on improving estimation accuracy using ensemble learning, specifically soft voting techniques (Voting)

II. Related Concepts

- **Single Models:**

KNN: A simple, distance-based baseline model.

Extra Trees: A strong, randomized tree-based ensemble.

Gradient Boosting: A sequential model that corrects previous errors.

Logistic Regression: A linear model used for baseline comparison.

- **Ensemble Learning (Soft Voting)**

Combining multiple individual models to make a final prediction, aiming to leverage the strengths of each algorithm.

II. Related Concepts

Datasets Evaluated:

- Desharnais
- Finnish
- ISBSG10

Evaluation Metrics:

- MAE & RMSE: Measure the error rate (Lower is better).
- MSE: Measures average squared difference (Lower is better).
- R^2 (R-squared): Measures how well the model explains the variance (Higher is closer to 1.0 is better).

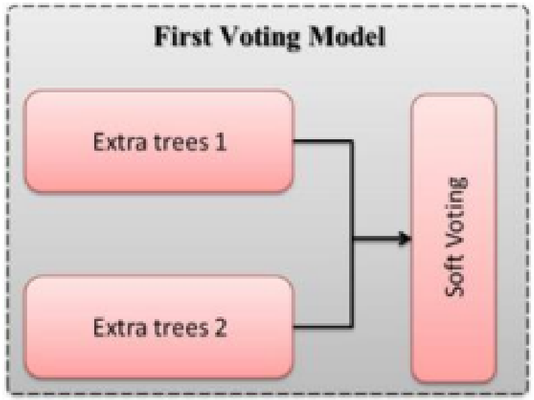
III. The Voting Ensemble Architectures

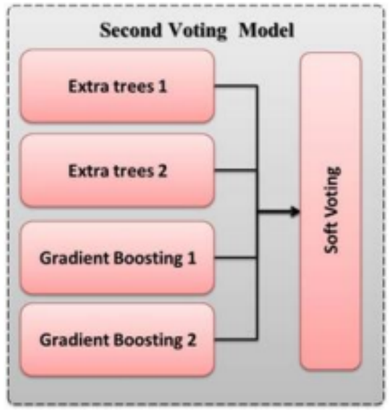
Voting 1: 2 × Extra Trees

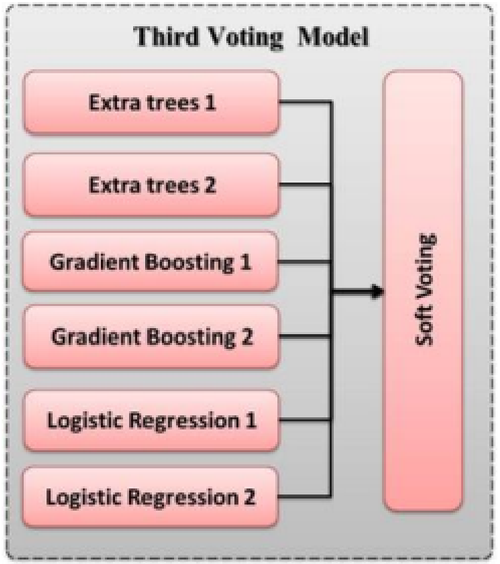
Voting 2: 2 × Extra Trees + 2 × Gradient Boosting

Voting 3: 2 × Extra Trees + 2 × Gradient Boosting + 2 × Logistic Regression

Core Approach: Utilizing soft voting to blend heterogeneous models (tree-based and linear) for balanced predictions







IV. Technical Implementation

Environment: Python ecosystem (Scikit-learn).

Pipeline Steps:

- Load and clean benchmark datasets.
- Standardize data for model stability.
- Train/Test split (e.g., 80/20).
- Train single models and voting ensembles.
- Evaluate using standard regression metrics.

V. Technical Implementation

These results are from the original paper, used for comparison

Dataset	Best Single Model	R^2 (Single)	Best Voting Model	R^2 (Voting)	Improvement
Desharnais	Extra Trees	6.937	Voting 3	**7.856**	⬆ +0.0919
Finnish	KNN	9.943	Voting 2	**9.990**	⬆ +0.0047
ISBSG10	Extra Trees	8.536	Voting 1	**8.189**	⬇ -0.0347

- Ensemble models generally improve performance, but not always
- Best voting model varies by dataset
- Increasing model complexity does not guarantee better results
- Dataset characteristics strongly influence model effectiveness

V. Technical Implementation

Dataset 1: Desharnais

Model	MAE ↓	RMSE ↓	R^2 ↑	Rank
KNN	5.098	6.667	0.2532	Worst
Extra Trees	4.379	5.855	0.4240	Good
Voting 1	4.343	5.777	0.4393	Better
Voting 2	**3.981**	5.465	0.4982	**Best**
Voting 3	3.996	**5.461**	**0.4989**	**Very good**

V. Technical Implementation

Dataset 2: Finnish

Model	MAE ↓	RMSE ↓	R^2 ↑	Rank
KNN	15.191.885	19.592.337	0.4448	Worst
Extra Trees	13.329.551	17.599.676	0.5520	Good
Voting 1	13.312.210	17.530.851	0.5555	Better
Voting 2	14.007.501	18.413.977	0.5096	Worse
Voting 3	**13.199.935**	**16.808.858**	**0.5913**	**Best**

V. Technical Implementation

Dataset 3: ISBSG10

Model	MAE ↓	RMSE ↓	R^2 ↑	Rank
KNN	78.386.380	128.429.418	0.3116	Worst
Extra Trees	66.398.548	111.540.776	0.4808	Good
Voting 1	67.031.926	112.471.062	0.4721	Slightly worse
Voting 2	68.552.576	112.615.850	0.4707	Worse
Voting 3	**63.113.779**	**101.700.633**	**0.5683**	**Best**

V. Technical Implementation

Analysis of Experimental Results

- Ensemble models outperform single models across all datasets
- Extra Trees is a strong baseline, while KNN performs worst
- Voting models improve accuracy by combining multiple algorithms
- **Voting 3** often performs best due to **model diversity** (tree-based + boosting + linear)
- However, more models do not always guarantee better performance

→ effectiveness depends on dataset characteristics

- Performance differences between voting models are small

→ models are stable and consistent

- **No universal best model**

→ model selection depends on the dataset

V. Technical Implementation

Key Findings:

- Ensemble models generally outperform single base models.
- KNN often struggles as a baseline, while Extra Trees provides strong, stable individual results.
- Voting models successfully improve overall prediction accuracy and stability.

The "Best" Model Paradigm:

- There is no single "universal best" model.
- The best voting model varies depending on the dataset
- Works well across multiple datasets

Important Insight:

- Model performance is heavily dependent on specific dataset characteristics.
- More models ≠ always better. Compatibility and model diversity are the true key factors.

VI. Advantages & Limitations

Strengths:

- Effectively uses soft voting to enhance accuracy.
- Improves generalizability by testing on multiple datasets.
- Uses comprehensive evaluation metrics.

Limitations:

- Lacks advanced techniques like stacking or deeper boosting.
- Base models lack hyperparameter tuning.
- Linear models (Logistic Regression) may reduce performance on nonlinear data.
- Validated on limited dataset sizes

VII. Directions for Future Research

- Explore advanced ensemble methods, such as Stacking algorithms.
- Perform systematic hyperparameter tuning for base models to unlock maximum accuracy.
- Validate the proposed models on larger, more complex real-world datasets.
- Investigate the application of Deep Learning architectures.

VIII. Conclusion

- Ensemble learning, specifically soft voting, is a highly effective approach for software effort estimation.
- Voting models generally outperform or stabilize the predictions of single baseline models.
- There is no universal best model; performance depends on the dataset.
- Performance differences between Voting models are often small.

Scan the QR code to view the demo source code

THANK YOU FOR LISTENING!